YOUTH ENTREPRENEURS CHALLENGE 2027

FUSION ENGINE TECHNOLOGY

INNOVATE • INTEGRATE • ACCELERATE

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PROJECT SUBMISSION REPORT

Innovation Title: EvacAccess India — Accessibility-Aware Emergency Evacuation & Decision-Support System

Theme Category: Accessibility & Inclusion | **Portal:** MY Bharat (YEC 2027)

1. Executive Summary

Fusion Engine Technology is a GovTech and civic assistive engineering venture dedicated to delivering high-reliability public safety software. Our flagship platform, **EvacAccess India**, addresses a life-critical blindspot in Indian disaster response logistics: the safe and dignified evacuation of persons with disabilities (Divyangjan), senior citizens, and patients dependent on electric life-support machinery (oxygen concentrators and ventilators).

During floods and cyclones, standard navigation applications direct evacuees through flooded streets or toward facilities lacking wheelchair ramps and continuous power backup. EvacAccess India provides an explainable, deterministic spatial decision engine that cross-references live flood inundation, micro-topographical barriers, and audited shelter capabilities to guarantee verified safe corridors.

2. The Core Problem Statement

- **The Navigation Blindspot:** Mainstream routing tools optimize solely for motor vehicle travel time and distance, failing to identify stairs, high curbs, or 15cm floodwaters that immediately stall electric wheelchairs.
- **Shelter Infrastructure Mismatch:** Emergency shelters are marked as active in central directories without verifying ramp slope (1:12), wide restroom doors, or dedicated generator circuits for ventilators.
- **Disproportionate Human Vulnerability:** Across India, over 2.68 crore persons with disabilities and over 14 crore senior citizens experience **2x to 4x higher casualty rates** during natural calamities due to non-inclusive disaster logistics.

3. Technical Architecture & Innovation

EvacAccess India integrates a three-tier government-grade software ecosystem:

A. Deterministic Spatial Feasibility Engine

Unlike opaque black-box AI algorithms, EvacAccess India evaluates candidate corridors deterministically against individual citizen requirements and outputs three transparent outcomes:

- `FEASIBLE_WITH_AVAILABLE_DATA`: All mandatory accessibility constraints verified (ramp verified, step-free corridor, active 24/7 generator circuit, and zero water obstacles).
- `UNSUITABLE`: Confirmed physical barrier detected (e.g., stairs on corridor, shelter without ramp, flood depth >10cm).
- `UNVERIFIED`: Hazard data older than 60 minutes or shelter inspection missing.

B. Tactical GIS Web Command Center

Built for State Disaster Management Authorities (SDMAs) and District Emergency Operations Centers (DEOCs). Employs CartoDB Positron daylight cartography with real-time flood polygons, shelter capacity gauges, and five role-based operational consoles (Incident Commander, Field Unit, Safety Reviewer, System Admin, and Simulation Observer).

C. Offline-First Mobile Field Application

Cross-platform Flutter application compiled for Android (`app-debug.apk`, 92.4 MB). Features local SQLite storage and cryptographic idempotency, enabling NDRF field teams to conduct shelter inspections and report road barriers completely offline during network blackouts.

D. Accessible Multilingual Alert Studio

Generates emergency broadcast notices in English, Hindi, and regional languages with WCAG 2.1 AAA high-contrast modes and text-to-speech audio synthesis.

4. Current Traction & Prototype Validation

- **Technology Readiness Level (TRL): TRL 5–6** (Integrated Prototype Tested in Relevant Simulation Environment).
- **Automated Test Suite:** 100% test pass rate (`pytest 7/7 passed`) across permission models, routing barrier evaluations, and cryptographic audit records.
- **Clean Production Build:** Frontend dashboard compiled with zero errors (`pnpm build` verified); native Android APK compiled and ready for deployment.
- **Field Simulation:** Successfully validated against scenario **EA-1001** (Puri Coastal Inundation Emergency, Odisha).

5. National Policy Alignment & Social Impact

- **Rights of Persons with Disabilities (RPwD) Act 2016 (Section 8):** Directly satisfies statutory mandates for equal safety and protection of persons with disabilities in situations of disaster.
- **National Disaster Management Plan (NDMP 2019):** Implements social inclusion and vulnerability mapping guidelines.
- **Sugamya Bharat Abhiyan (Accessible India Campaign):** Extends physical accessibility standards into digital emergency evacuation logistics.
- **Global Alignment:** Priority 4 of the UN Sendai Framework for Disaster Risk Reduction and UN Sustainable Development Goals (SDG 10 & 11).

6. Business & Deployment Model (B2G)

PHASE	TARGET AGENCIES	DEPLOYMENT SCOPE
Phase 1 (Months 1–6)	District Disaster Management Authorities (DDMAs)	Pilot deployments in coastal flood zones (Odisha, Tamil Nadu).
Phase 2 (Months 6–18)	State Disaster Management Authorities (SDMAs)	State-level annual software licensing & maintenance contracts (AMC).
Phase 3 (Months 18–36)	ERSS 112 India & Smart City ICCCs	Nationwide API integration with emergency response networks.

7. Corporate Profile & Contact Information

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Theme Category	Accessibility & Inclusion
Event / Portal	Youth Entrepreneurs Challenge (YEC) 2027 — MY Bharat